```
!pip install -U transformers datasets accelerate flask flask-cors pyngrok
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Collecting transformers
  Downloading transformers-5.3.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting datasets
  Downloading datasets-4.6.1-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.12.0)
Collecting accelerate
  Downloading accelerate-1.13.0-py3-none-any.whl.metadata (19 kB)
Requirement already satisfied: flask in /usr/local/lib/python3.12/dist-packages (3.1.3)
Collecting flask-cors
  Downloading flask_cors-6.0.2-py3-none-any.whl.metadata (5.3 kB)
Collecting pyngrok
  Downloading pyngrok-7.5.0-py3-none-any.whl.metadata (8.1 kB)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.24.3)
Collecting pyarrow>=21.0.0 (from datasets)
  Downloading pyarrow-23.0.1-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (3.1 kB)
Requirement already satisfied: dill<0.4.1,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: httpx<1.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.28.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.19 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2026.2.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: torch>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from accelerate) (2.10.0+cu128)
Requirement already satisfied: blinker>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in /usr/local/lib/python3.12/dist-packages (from flask) (8.3.1)
Requirement already satisfied: itsdangerous>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in /usr/local/lib/python3.12/dist-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from flask) (3.1.6)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (4.12.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (1.0.
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->datasets) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0.0->data
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datas
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (75.2
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate) (1
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accelerate)
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0.0->accele
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch>=2.0
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch>=2
```

```
import pandas as pd
import torch
from sklearn.model_selection import train_test_split
from datasets import Dataset
from transformers import DistilBertTokenizerFast
from transformers import DistilBertForSequenceClassification
from transformers import TrainingArguments
from transformers import Trainer
```

```
df = pd.read_csv("/content/train.csv")

df["comment_text"] = df["comment_text"].fillna("").astype(str)

label_cols = [
    "toxic","severe_toxic","obscene",
    "threat","insult","identity_hate"
]

print("Dataset size:", len(df))
```

```
Dataset size: 159571
```

```
train_df, val_df = train_test_split(df, test_size=0.1, random_state=42)

train_dataset = Dataset.from_pandas(train_df)
val_dataset = Dataset.from_pandas(val_df)
```

```
tokenizer = DistilBertTokenizerFast.from_pretrained(
    "distilbert-base-uncased"
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 2.24kB/s]

vocab.txt: 232k/? [00:00<00:00, 3.22MB/s]

tokenizer.json: 466k/? [00:00<00:00, 3.98MB/s]

```
def tokenize(example):

    return tokenizer(
        example["comment_text"],
        padding="max_length",
        truncation=True,
        max_length=256
    )

train_dataset = train_dataset.map(tokenize, batched=True)
val_dataset = val_dataset.map(tokenize, batched=True)
```

Map: 100% 143613/143613 [00:58<00:00, 3182.58 examples/s]

Map: 100% 15958/15958 [00:08<00:00, 1377.95 examples/s]

```
def format_labels(example):

    example["labels"] = [
        float(example["toxic"]),
        float(example["severe_toxic"]),
        float(example["obscene"]),
        float(example["threat"]),
        float(example["insult"]),
        float(example["identity_hate"])
    ]

    return example


train_dataset = train_dataset.map(format_labels)
val_dataset = val_dataset.map(format_labels)
```

Map: 100% 143613/143613 [00:20<00:00, 5704.57 examples/s]

Map: 100% 15958/15958 [00:02<00:00, 9522.63 examples/s]

```
train_dataset.set_format(
    type="torch",
    columns=["input_ids","attention_mask","labels"]
)

val_dataset.set_format(
    type="torch",
    columns=["input_ids","attention_mask","labels"]
)
```

```
model = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=6,
    problem_type="multi_label_classification"
)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
```

config.json: 100% 483/483 [00:00<00:00, 12.2kB/s]

model.safetensors: 100% 268M/268M [00:03<00:00, 309MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 3585.27it/s]

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.bias   | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
classifier.bias         | MISSING    |
classifier.weight       | MISSING    |
pre_classifier.bias     | MISSING    |
pre_classifier.weight   | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream t
```

```
training_args = TrainingArguments(

    output_dir="./results",

    learning_rate=2e-5,

    per_device_train_batch_size=8,

    per_device_eval_batch_size=8,

    num_train_epochs=2,

    weight_decay=0.01,

    logging_dir="./logs",

    fp16=True
)
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
```

```
trainer = Trainer(

    model=model,

    args=training_args,

    train_dataset=train_dataset,

    eval_dataset=val_dataset
)
```

```
trainer.train()
```

[24000/30004 49:38, Epoch 2/2]

Step	Training Loss
500	0.115283
1000	0.060851
1500	0.053177
2000	0.051774
2500	0.050534
3000	0.051597
3500	0.043706
4000	0.050031
4500	0.044999
5000	0.045930
5500	0.046138
6000	0.040624
6500	0.044619
7000	0.044573
7500	0.045432
8000	0.040518
8500	0.041595
9000	0.043775
9500	0.043229
10000	0.042822
10500	0.041689
11000	0.038213
11500	0.042950
12000	0.039839
12500	0.044232
13000	0.039994
13500	0.039013
14000	0.039047
14500	0.040227
15000	0.037016
15500	0.039866
16000	0.036826
16500	0.039425
17000	0.038193
17500	0.036893
18000	0.035815
18500	0.033064
19000	0.035162
19500	0.028773
20000	0.033629
20500	0.029227
21000	0.033608
21500	0.026658
22000	0.033277
22500	0.034769
23000	0.034676
23500	0.033444
24000	0.031182

24500	0.031256
25000	0.029600
25500	0.031651
26000	0.032928
26500	0.030467
27000	0.027461
27500	0.031133
28000	0.030640
28500	0.026914
29000	0.030060
29500	0.031928
30000	0.028104
30500	0.032292
31000	0.029839
31500	0.025440
32000	0.028098
32500	0.028714
33000	0.028637
33500	0.029802
34000	0.031393
34500	0.028500
35000	0.028488
35500	0.028730

Writing model shards: 100%	1/1 [00:00<00:00, 1.58it/s]
Writing model shards: 100%	1/1 [00:06<00:00, 6.98s/it]
Writing model shards: 100%	1/1 [00:04<00:00, 4.34s/it]
Writing model shards: 100%	1/1 [00:04<00:00, 4.87s/it]
Writing model shards: 100%	1/1 [00:01<00:00, 1.09s/it]
Writing model shards: 100%	1/1 [00:02<00:00, 2.87s/it]
Writing model shards: 100%	1/1 [00:02<00:00, 2.93s/it]
Writing model shards: 100%	1/1 [00:02<00:00, 2.07s/it]
Writing model shards: 100%	1/1 [00:00<00:00, 1.03it/s]
Writing model shards: 100%	1/1 [00:01<00:00, 1.16s/it]
Writing model shards: 100%	1/1 [00:00<00:00, 1.01it/s]
Writing model shards: 100%	1/1 [00:05<00:00, 5.99s/it]
Writing model shards: 100%	1/1 [00:01<00:00, 1.10s/it]
Writing model shards: 100%	1/1 [00:01<00:00, 1.42s/it]
Writing model shards: 100%	1/1 [00:05<00:00, 5.20s/it]
Writing model shards: 100%	1/1 [00:06<00:00, 6.13s/it]
Writing model shards: 100%	1/1 [00:07<00:00, 7.15s/it]
Writing model shards: 100%	1/1 [00:04<00:00, 4.19s/it]
Writing model shards: 100%	1/1 [00:01<00:00, 1.02s/it]
Writing model shards: 100%	1/1 [00:07<00:00, 7.20s/it]
Writing model shards: 100%	1/1 [00:05<00:00, 5.73s/it]
Writing model shards: 100%	1/1 [00:05<00:00, 5.23s/it]
Writing model shards: 100%	1/1 [00:00<00:00, 1.02it/s]
Writing model shards: 100%	1/1 [00:00<00:00, 1.16it/s]
Writing model shards: 100%	1/1 [00:04<00:00, 4.63s/it]
Writing model shards: 100%	1/1 [00:00<00:00, 1.19it/s]
Writing model shards: 100%	1/1 [00:04<00:00, 4.55s/it]

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Writing model shards: 100%                                                1/1 [00:01<00:00,  1.30s/it]
```

```
model.save_pretrained("toxicity_model")

tokenizer.save_pretrained("toxicity_model")
```

```
Writing model shards: 100%                                                1/1 [00:00<00:00,  1.18it/s]
Writing model shards: 100%                                                1/1 [00:01<00:00,  1.37s/it]
Writing model shards: 100%                                                1/1 [00:00<00:00,  1.19it/s]
('toxicity_model/tokenizer_config.json', 'toxicity_model/tokenizer.json')
```

```
Writing model shards: 100%                                                1/1 [00:00<00:00,  1.19it/s]
```

```
from flask import Flask, request, jsonify
from flask_cors import CORS
import torch

from transformers import DistilBertTokenizerFast
from transformers import DistilBertForSequenceClassification

app = Flask(__name__)
CORS(app)

tokenizer = DistilBertTokenizerFast.from_pretrained("toxicity_model")

model = DistilBertForSequenceClassification.from_pretrained("toxicity_model")

model.eval()

labels = [
    "toxic",
    "severe_toxic",
    "obscene",
    "threat",
    "insult",
    "identity_hate"
]

@app.route("/predict", methods=["POST"])
def predict():

    data = request.get_json()

    text = data["comment"]

    inputs = tokenizer(
        text,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=256
    )

    with torch.no_grad():
        outputs = model(**inputs)

    probs = torch.sigmoid(outputs.logits)[0].tolist()

    result = {
        labels[i]: float(probs[i]) for i in range(len(labels))
    }

    result["is_toxic"] = any(p > 0.6 for p in probs)

    return jsonify(result)
```

```
Writing model shards: 100%                                                1/1 [00:01<00:00,  1.63s/it]
Loading weights: 100%                                                104/104 [00:00<00:00, 2501.66it/s]
Writing model shards: 100%                                                1/1 [00:04<00:00,  4.34s/it]
```

```
from pyngrok import ngrok

ngrok.set_auth_token("30Ey05TtMXSlFGyVDDe6SmGAe2E_2QCkRMCn6SHqBkkm7ABsM")

public_url = ngrok.connect(5000)

print("Public URL:", public_url)
```

```
TrainOutput(global_step=25904, training_loss=0.037916534804405 38, metrics={'train_runtime': 2997.0309,
'train_samples_per_second': 95.837, 'train_steps_per_second': 11.98, 'total_flos': 1.9025397590206464e+16, 'train_loss':
0.037916534804405 38, 'epoch': 2.0})
ERROR:root:Unexpected exception finding object.shape
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/google/colab/_debugpy_repr.py", line 54, in get_shape
    shape = getattr(obj, 'shape', None)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/werkzeug/local.py", line 318, in __get__
```